Weiji Xie

M.S. student (Ph.D. track), Shanghai Jiao Tong University
Motion control and reinforcement learning
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Personal Website

Seeking a research internship in dexterous manipulation, embodied agents, or motion control. Available spring–fall 2027 (J-1 intern visa for the US); in China from Dec 2026.
11 papers, **5** as first / co-first author (NeurIPS, CoRL, IROS) · **260+** citations · **1.7k+** GitHub stars (PBHC, TextOp, TTC)

Education

Shanghai Jiao Tong University — M.S. in Computer Science and Technology — Sept 2025 – present
Advisor: Prof. Weinan Zhang. Research: reinforcement learning and humanoid robot control.

Shanghai Jiao Tong University — B.Eng. in Artificial Intelligence — Sept 2021 – June 2025
GPA 4.07 / 4.30, ranked in the top 5%.

Research Experience

TeleAI, China Telecom — Research Intern, Humanoid Robot Learning — Shanghai | Oct 2024 – Present
Mentor: Dr. Chenjia Bai. Reinforcement learning for humanoid whole-body control; vision-language-action (VLA) models for loco-manipulation.

APEX Lab, Shanghai Jiao Tong University — Undergraduate Researcher — Shanghai | Aug 2022 – June 2025
Advisor: Prof. Weinan Zhang. Multi-agent reinforcement learning and legged robot locomotion.

Projects

Motion Tracking for Highly Dynamic Humanoid Whole-Body Control

- Built the pipeline from human video to the real robot: motion extraction, physics-aware filtering and retargeting, RL motion tracking with an error-adaptive curriculum, and sim-to-real deployment, **enabling highly dynamic motions such as spinning and aerial kicks on the Unitree G1** (KungfuBot, NeurIPS 2025, first author).
- Extended tracking to many skills under a single policy, accurate in both body pose and global trajectory: minute-long martial-arts and dance sequences, and motions unseen in training (KungfuBot2, ICRA 2026).
- Core maintainer of the open-source project PBHC: code for the full pipeline and a developer community, **1,000+ GitHub stars**.

Language- and Vision-Driven Humanoid Whole-Body Control and Loco-Manipulation

- Built a real-time text-driven motion generation and control framework: an autoregressive motion-diffusion model turns streaming text commands into trajectories, and a whole-body tracking policy executes them on the real robot. Commands can change mid-motion; dancing, jumping and other skills chain without stopping (TextOp, CoRL 2026, first author; **550+ GitHub stars**).
- Proposed test-time command optimization for humanoid behavior foundation models (BFMs) to address controller-dependent demonstration data for VLA training and insufficient BFM execution accuracy in manipulation tasks. Optimized input motion commands through simulation search to improve execution accuracy without retraining the controller, narrowing the gap between commands and execution. Derived a computation allocation strategy that is approximately compute-optimal, and enabled demonstrations to be converted into training data for different controllers (TTC, first author).
- Contributed to multi-source VLA training for whole-body loco-manipulation: data pipelines for real and simulated teleoperation, egocentric human video and simulation augmentation; VLA policies for real-robot pick-and-place, carrying and wiping. Egocentric human demonstrations raised success by **45+ points** on conditions absent from robot data (KODA, OASIS).

Humanoid Locomotion and Teleoperation on Complex Terrain

- Combined dynamic-balance mechanisms with reinforcement learning, using only proprioceptive state to achieve stable walking through narrow passages and over obstacle terrain on the Unitree H1-2; validated balance recovery under external disturbances (DBHL, IROS 2025, first author).
- Contributed to perceptive whole-body teleoperation that fuses human motion commands, depth-based terrain perception and robot state: the robot adapts its posture and contacts on stairs and slopes while the operator moves on flat ground (NEXUS).

Publications

* equal contribution † corresponding author

TextOp: Real-time Interactive Text-Driven Humanoid Robot Motion Generation and Control — **CoRL 2026**
Weiji Xie*, Jiakun Zheng*, Jinrui Han, Jiyuan Shi, Weinan Zhang†, Chenjia Bai†, Xuelong Li
project arXiv code

KungfuBot: Physics-Based Humanoid Whole-Body Control for Learning Highly-Dynamic Skills — **NeurIPS 2025**
Weiji Xie*, Jinrui Han*, Jiakun Zheng*, Huanyu Li, Xinzhe Liu, Jiyuan Shi, Weinan Zhang, Chenjia Bai†, Xuelong Li† — project arXiv code

KungfuBot2: Learning Versatile Motion Skills for Humanoid Whole-Body Control — **ICRA 2026**
Jinrui Han, **Weiji Xie**, Jiakun Zheng, Jiyuan Shi, Weinan Zhang, Ting Xiao, Chenjia Bai — project arXiv

Humanoid Whole-Body Locomotion on Narrow Terrain via Dynamic Balance and Reinforcement Learning — **IROS 2025**
Weiji Xie, Chenjia Bai†, Jiyuan Shi, Junkai Yang, Yunfei Ge, Weinan Zhang†, Xuelong Li — project arXiv

Test-Time Compute for Humanoid Behavior Foundation Models — **Under Review**
Weiji Xie*, Junsong Wu*, Zibo Zhou, Yueru Chen, Xingyi Wang, Zhenhua Xiong, Weinan Zhang, Chenjia Bai† — project arXiv code

HALO: Closing Sim-to-Real Gap for Heavy-loaded Humanoid Agile Motion Skills via Differentiable Simulation — **IROS 2026**
Xingyi Wang*, Chenyun Zhang*, **Weiji Xie***, Chao Yu, Wei Song, Chenjia Bai†, Shiqiang Zhu† — project arXiv

KODA: Kinematics–Observation Data Alignment for Humanoid Learning from Human Egocentric Demonstration — **Under Review**
Zibo Zhou*, Yueru Chen*, Junsong Wu, **Weiji Xie**, Qingyao Xu, Sheng Yin, Shiqing Wang, Jiyuan Shi, Siheng Chen, Chenjia Bai†, Xuelong Li† — project arXiv

NEXUS: Perceptive Whole-Body Control for Terrain-Adaptive Teleoperation — **Under Review**
Xiangyu Miao, Junsong Wu, Jiyuan Shi, **Weiji Xie**, Jinrui Han, Xingyi Wang, Weinan Zhang†, Chenjia Bai†, Xuelong Li — project arXiv

LoopSR: Looping Sim-and-Real for Lifelong Policy Adaptation of Legged Robots — **IROS 2025**
Peilin Wu, **Weiji Xie**, Jiahang Cao, Hang Lai, Weinan Zhang — arXiv

Offline Fictitious Self-Play for Competitive Games — **AAAI 2026**
Jingxiao Chen, **Weiji Xie**, Weinan Zhang†, Yong Yu, Ying Wen† — arXiv

OASIS: From Simulation Data Collection to Real-World Humanoid Loco-Manipulation — **Under Review**
Zehao Yu, Jiakun Zheng, **Weiji Xie**, Jiyuan Shi, Chenyun Zhang, Chenjia Bai, Xuelong Li — project arXiv

Skills

Robot learning	Reinforcement learning, motion retargeting, sim-to-real and system identification, diffusion motion generation, VLA training, sampling-based optimization
Robots & simulators	Unitree G1, H1-2, Sharpa Wave; Isaac Gym / Sim / Lab, MuJoCo / MJX
Engineering	Python (PyTorch, JAX), C++, CUDA, ROS, Linux, Git, Claude Code, Codex

Honors & Service

- ASC Student Supercomputer Challenge 2024: First Prize and Application Innovation Award
- SJTU Academic Excellence Scholarship (2022, 2023, 2024); Fuguang Undergraduate Scholarship (2021–2024)
- Outstanding Communist Youth League Member, Shanghai Jiao Tong University (2023)
- Reviewer: CoRL, ICRA, ICLR (conferences); TPAMI, RA-L (journals)